AUSELIA

ADS7057 test board

How a perfboard signal becomes SPI samples, one step at a time.

The topology, the datasheet facts that matter, the routing of the board, and what each part is and where it sits.

Version 0.2 / October 15, 2026 / Nicolás Villegas

How to read this manual

Read the text first. Then look at the figure. The figure shows the same thing with numbers.

01 **One panel, one step**

A page holds four steps in a 2 × 2 grid. Each panel has a title, a text and a figure.

02 **Example tag**

A figure with this tag is a worked example. Its numbers come from the netlist, the board file or the simulation.

03 **Source**

The KiCad project in hw/, the simulations in sim/, the vendor datasheets and one real JLCPCB quote.

04 **Estimate**

A number marked ESTIMATE is not measured. Read it as a size, not as a promise.

05 **Names**

R16 is a reference designator. It is the same name in the schematic, the BOM and the silkscreen file.

06 **Order**

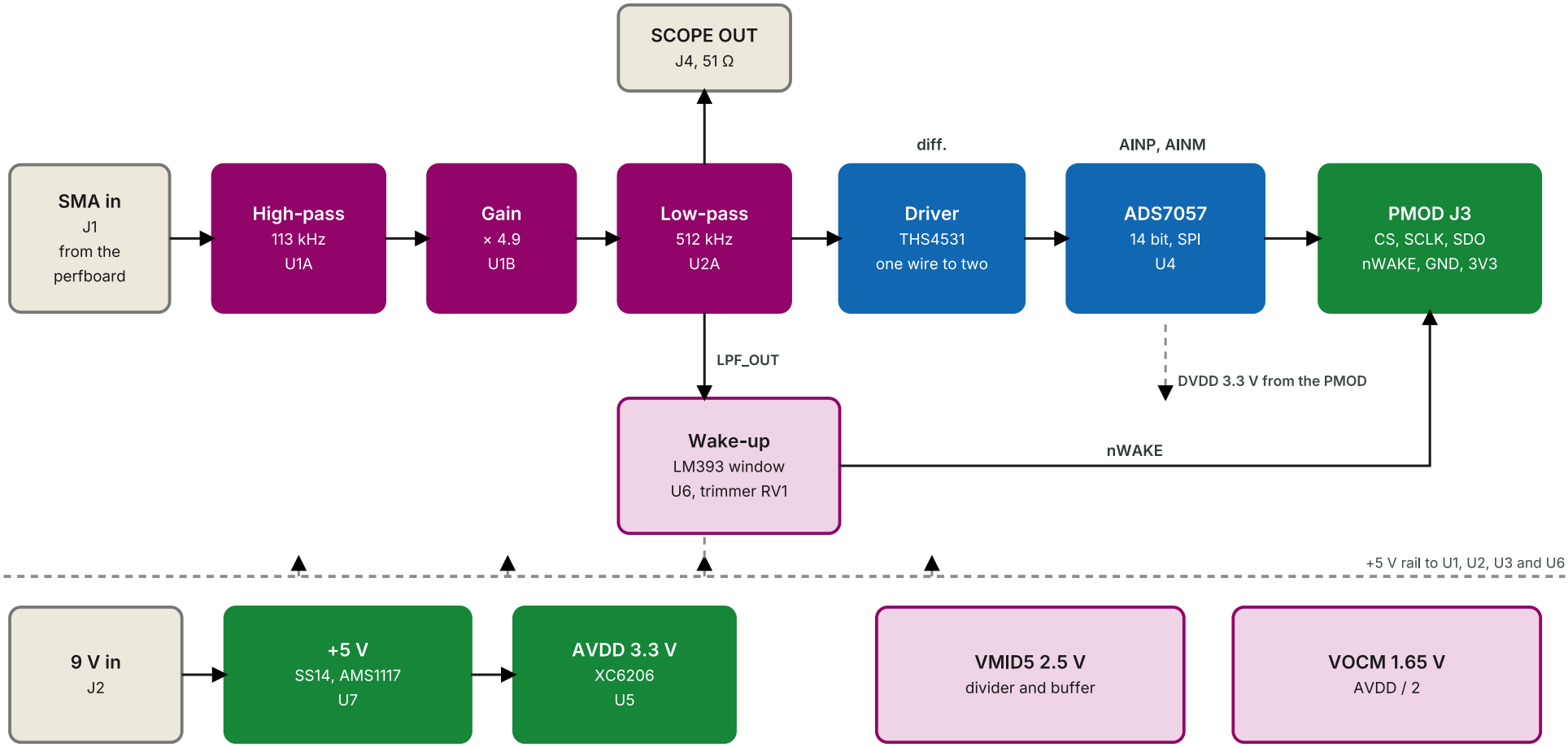
Part 0 gives the basics. Steps 1 to 54 follow the power, the signal, the ADC, SPI, the wake-up, the board and the bring-up.

Colors

signal chain · driver and ADC · power · control, data of the example · the item of this step

The board in one view

the signal goes left to right, the power runs along the bottom



THE BASICS

part 0

What the signal is, why the mid-rails, what an ADC measures and why the filter comes first.

POWER

steps 1 to 6

9 V to 5 V to 3.3 V, then the 2.5 V and 1.65 V mid-rails that the signal rides on.

SIGNAL AND ADC

steps 7 to 24

High-pass, gain, low-pass, then the driver and the ADC with its input filter.

SPI AND WAKE-UP

steps 25 to 35

The four SPI wires, the frame and its power-up rule, and the window comparator.

THE BOARD

steps 36 to 48

Layers, floor plan, every part and where it sits, routing, files for the factory and the cost.

BRING-UP

steps 49 to 54

Power-on checks, the first signal, the expected response and what is still open.

PART 0

The basics

0.1 TO 0.6

What the board is for, what the signal looks like, why there are two mid-rails, what an ADC measures, why the filter comes first and how to read a PCB. No board details yet.

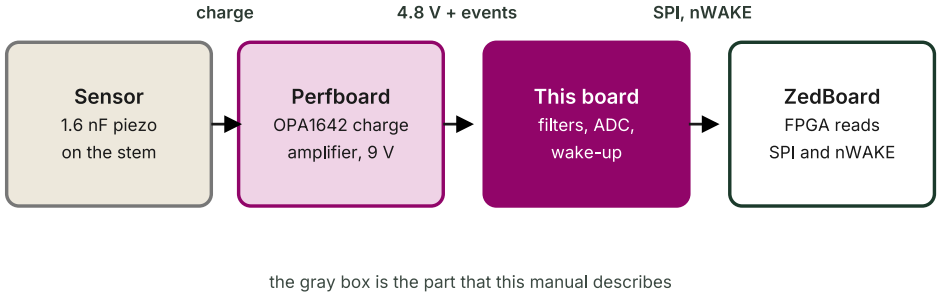
0.1 What the board is for

THE BASICS

A piezoelectric sensor on a plant stem hears the clicks of **acoustic emission** (AE). The perfboard charge amplifier turns the charge of the sensor into a voltage.

This board takes that voltage, filters it and measures it with a 14-bit ADC. It hands the numbers to the ZedBoard over SPI. It also raises a wake-up line when a click is strong.

CHAIN



the gray box is the part that this manual describes

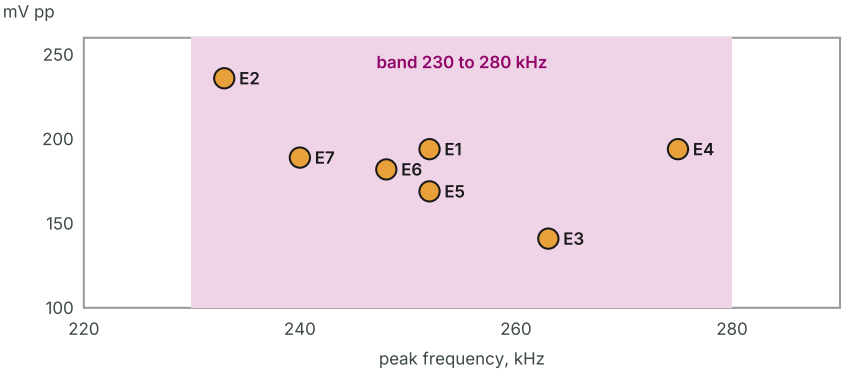
0.2 What the signal looks like

THE BASICS

The perfboard output rests near 4.8 V. An AE event rides on it as a burst between 233 and 275 kHz that lasts 26 to 83 µs. The seven recorded events are 141 to 236 mV peak to peak.

Between events, five of the eight baseline records show 3.7 to 4.6 mV rms of noise in the band. That noise sets the smallest event that this board can see.

7 RECORDED EVENTS



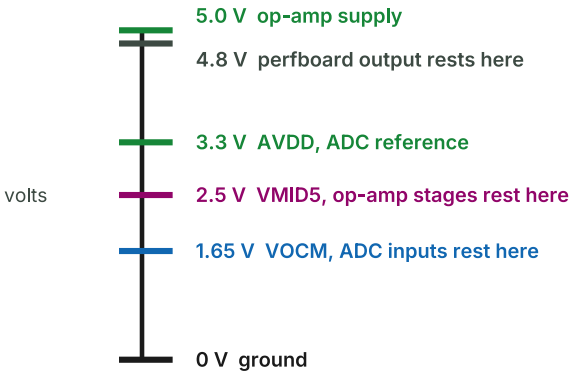
0.3 Why 2.5 V and 1.65 V

THE BASICS

This board has no negative supply. A signal that swings below 0 V would be cut off. So each stage rests at the middle of its own supply, and the signal swings around that level.

The op-amps run from 0 to 5 V, so they rest at **2.5 V**. The ADC runs from 0 to 3.3 V, so its inputs rest at **1.65 V**.

VOLTAGE LADDER



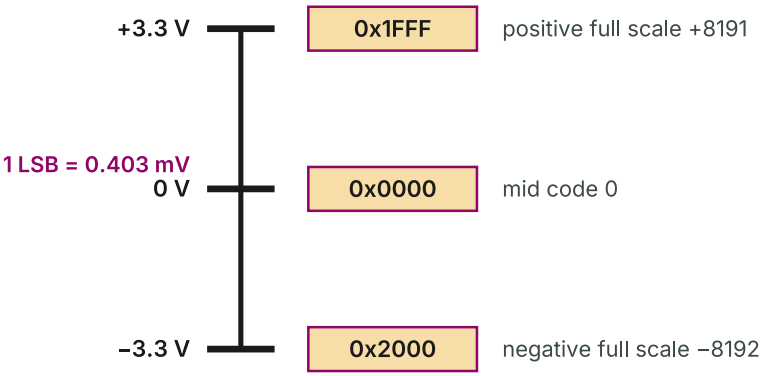
0.4 What an ADC measures

THE BASICS

The ADS7057 measures the **difference** between two pins, AINP and AINM. It reports the difference as a 14-bit signed number in two's complement. One step, called 1 LSB, is 0.403 mV.

The full-scale range is plus and minus the supply AVDD, so ±3.3 V. The code 0x0000 means no difference.

CODE LADDER



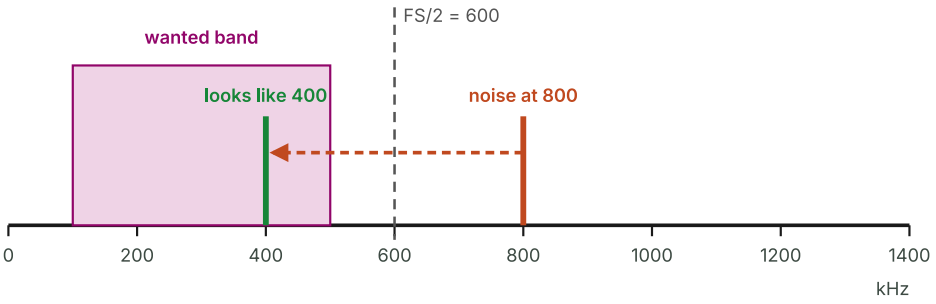
0.5 Why filter before sampling

THE BASICS

A sampled signal cannot tell a frequency f from FS – f. At 1.2 MSPS a tone at 800 kHz looks exactly like a tone at 400 kHz, inside the band of interest.

So a low-pass filter must remove what lies above half the sample rate before the ADC sees it. The sample rate here is about 1.2 MSPS with a 25 MHz SPI clock (ESTIMATE).

ALIASING



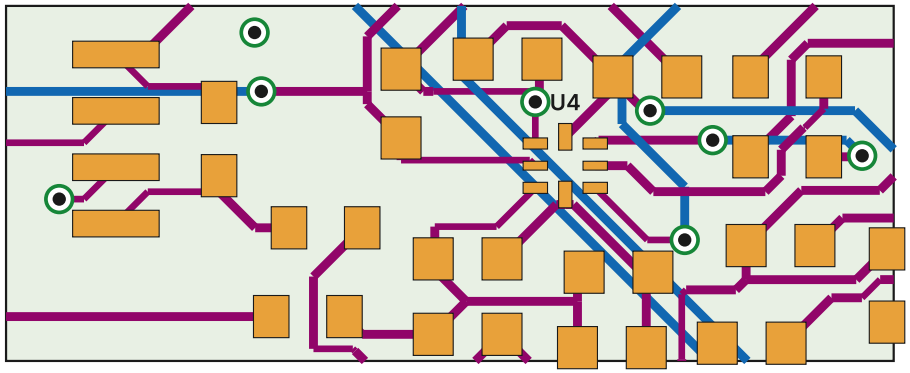
0.6 How to read a PCB

THE BASICS

The board has two copper layers. A **pad** holds a part. A **track** joins pads. A **via** is a plated hole that joins the two layers. A **pour** fills free space with copper, here ground.

The **silkscreen** prints the names. The figure is a 20 × 8 mm cut of this board around the ADC.

REAL BOARD, 6 ×



amber: pads purple: top tracks blue: bottom tracks green ring: vias pale green: ground pour

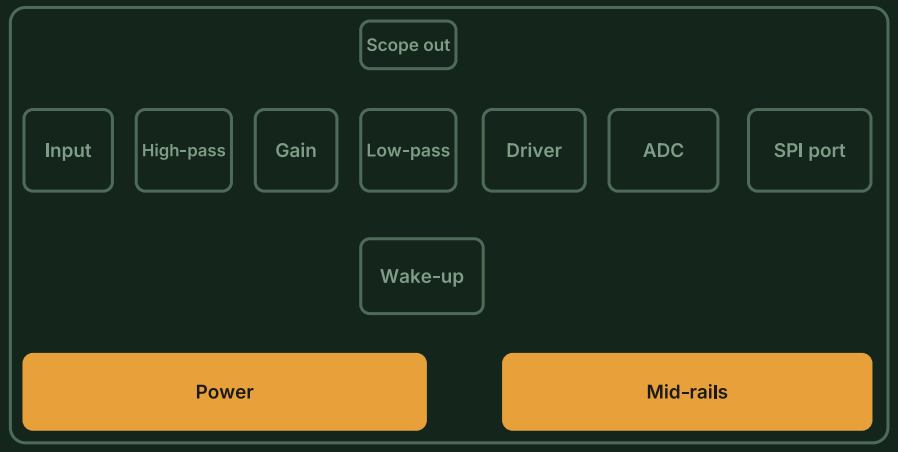
NOTES

NOTES

Power and references

STEPS 1 TO 6

How the 9 V battery becomes 5 V and 3.3 V, and how the two mid-rails of 2.5 V and 1.65 V are made. Everything else on the board rests on these rails.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

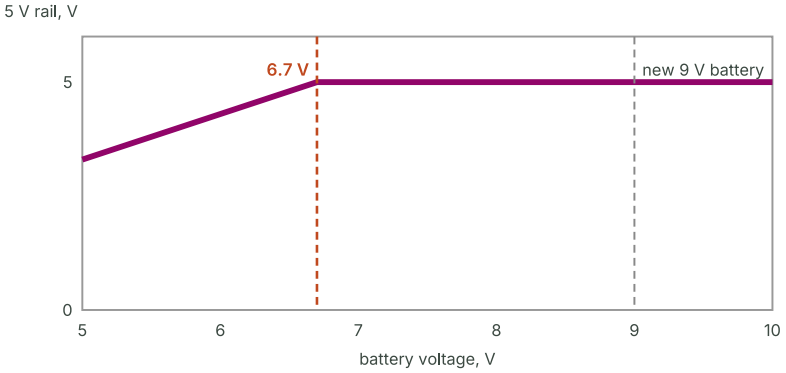
POWER AND REFERENCES

1 How 9 V becomes 5 V

A reversed battery would damage the board. The diode **SS14** (D1) blocks it and costs about 0.4 V. The regulator **AMS1117-5.0** (U7) then needs at least 1.3 V between its input and its output.

So the 5 V rail holds while the battery stays above 6.7 V. Below that, the rail follows the battery down.

5 V RAIL AGAINST THE BATTERY



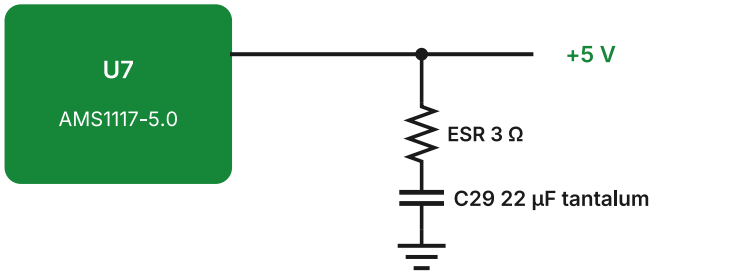
POWER AND REFERENCES

2 Why a tantalum capacitor on 5 V

The AMS1117 datasheet says the output capacitor is part of the regulator loop. It asks for 22 µF of solid tantalum for stability. It gives no ESR range.

A small ceramic has almost no series resistance, and some regulators of this type then oscillate. So C29 is a 22 µF tantalum with about 3 Ω of ESR at 100 kHz.

OUTPUT CAPACITOR



part LCSC C11366, 10 V rating on a 5 V rail

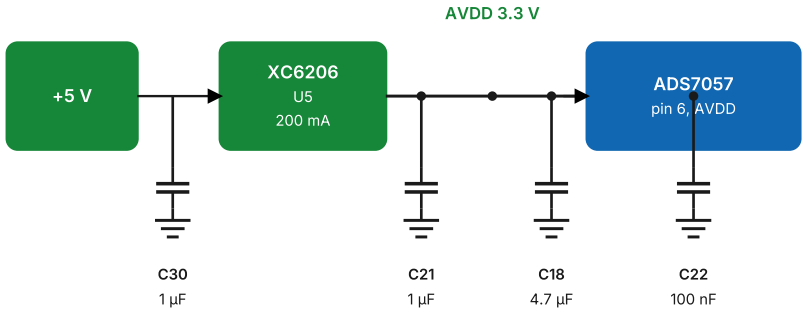
POWER AND REFERENCES

3 Where the ADC reference comes from

The ADS7057 uses its supply pin AVDD as its reference. Full scale is ±AVDD, so noise on AVDD is noise in every code.

A second regulator, the **XC6206** (U5), makes AVDD from the 5 V rail. It accepts ceramic capacitors, so the 1 µF parts C30 and C21 are enough. The 4.7 µF C18 and the 100 nF C22 sit at the ADC pin.

AVDD PATH



ESTIMATE: 0.5 mV rms of noise on AVDD adds about 0.2 LSB to an event of 17 % full scale

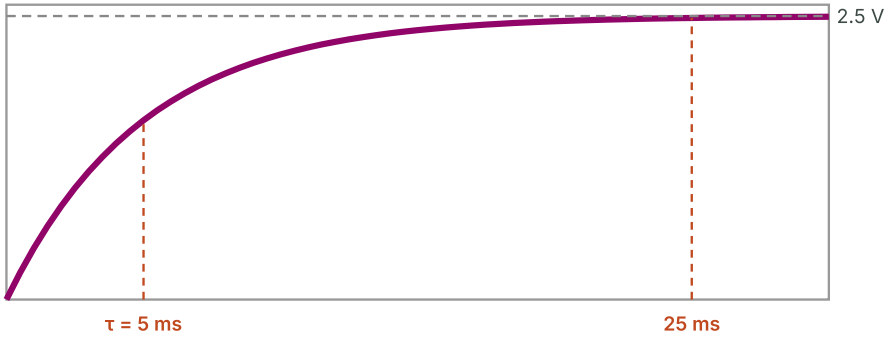
POWER AND REFERENCES

4 How the 2.5 V mid-rail is made

Two 10 kΩ resistors (R7, R8) divide the 5 V rail into 2.5 V. The 1 µF capacitor C7 smooths it. The op-amp U2B then copies this voltage and drives the rail called **VMID5**.

The divider and C7 form an RC of 5 ms, so VMID5 needs about 25 ms to settle after power-up. R9 (10 Ω) and C8 (10 µF) keep the buffer calm.

SETTLING AFTER POWER-UP



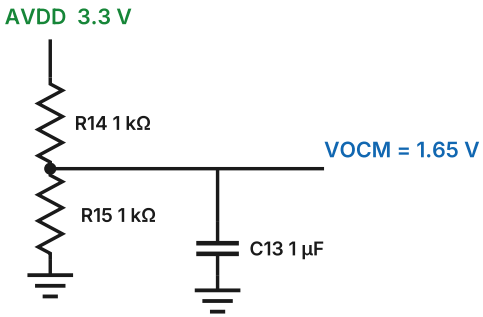
POWER AND REFERENCES

5 How the 1.65 V mid-rail is made

Two 1 kΩ resistors (R14, R15) divide **AVDD** in half, and C13 smooths the result. This voltage, VOCM, tells the THS4531 where to centre its two outputs.

The datasheet asks for the ADC inputs to rest at AVDD/2, within ±0.1 V. The divider follows AVDD by ratio, so any change of AVDD moves VOCM with it.

VOCM



allowed window

1.55 to 1.75 V

AVDD/2 ± 0.1 V

POWER AND REFERENCES

6 How much current, and where to probe

The board draws about 27 mA from the battery (ESTIMATE from datasheet typical values). The four op-amp channels are the largest part, and the AMS1117 uses 5 mA by itself.

At 8.6 V the AMS1117 dissipates about 80 mW. The pads of the capacitors below are the easiest probe points.

ESTIMATE, mA

- OPA2365, 4 channels 18
- AMS1117 own current 5
- VOCM divider 1.6
- ADS7057 AVDD 1.1
- LM393 0.4
- THS4531 0.25
- VMID5 divider 0.25
- threshold chain 0.065

+5 V	C29 pad 1	**AVDD**	C18 pad 1	**VMID5**	C8 pad 1
VOCM	C13 pad 1	**+3V3**	C20 pad 1	**GND**	J2 pin 2

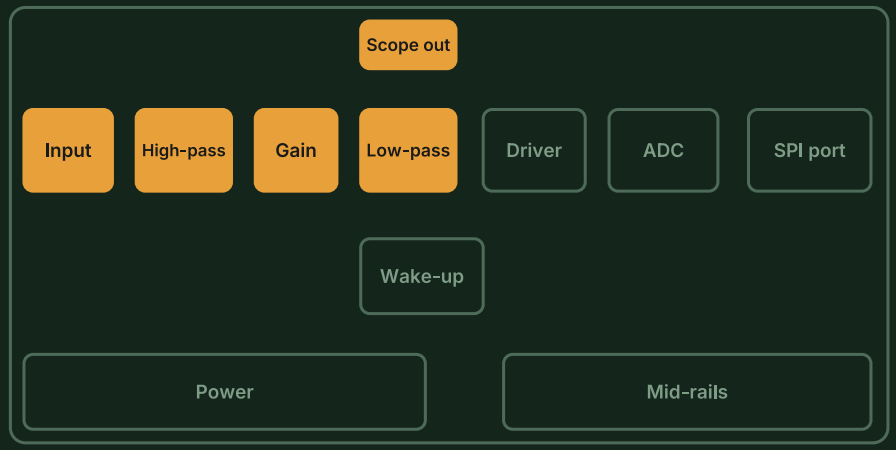
NOTES

NOTES

The signal path

STEPS 7 TO 16

From the SMA connector to the output of the low-pass filter: input coupling, high-pass, gain, low-pass, the total response, headroom, the SCOPE OUT tap and the op-amps.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

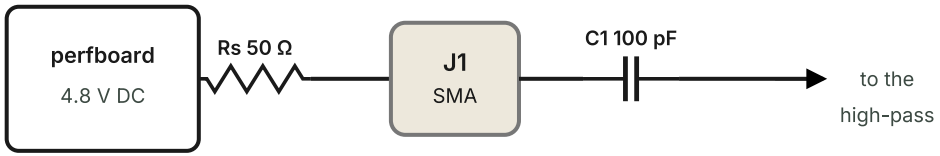
THE SIGNAL PATH

7 How the signal gets in

The perfboard output reaches J1 through a coax cable. Its own output resistor is 50 Ω. Its output rests at 4.8 V, which is far from the 2.5 V that this board needs.

So the first part is a 100 pF capacitor, C1. It passes the 250 kHz burst and blocks the DC level. A resistor to VMID5 later sets the new DC level.

INPUT



at 250 kHz, C1 is 6.4 kΩ. At DC it is open.

THE SIGNAL PATH

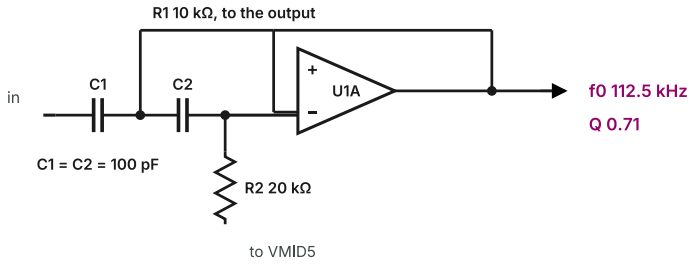
8 How the high-pass works

The high-pass is a **Sallen-Key** stage: two capacitors in series, two resistors and one op-amp used as a buffer. Low frequencies cannot pass the capacitors.

$$f_0 = \frac{1}{2\pi C\sqrt{R_1 R_2}}, \qquad Q = \frac{1}{2}\sqrt{\frac{R_2}{R_1}}$$

With C = 100 pF, R1 = 10 kΩ and R2 = 20 kΩ, the corner is 112.5 kHz. Q is 0.71. That is a maximally flat response.

SALLEN-KEY HIGH-PASS



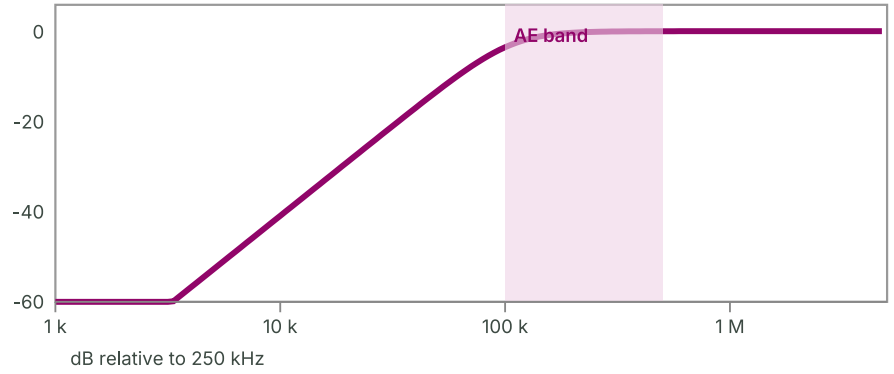
THE SIGNAL PATH

9 Why the corner is at 113 kHz

Most noise on the perfboard sits below 100 kHz, from 11.5 to 26.8 mV rms. The AE band starts at 100 kHz and holds only about 4 mV rms.

So the high-pass sits just under the band. At 10 kHz it already removes 41 dB of this stage alone. A corner any higher would also cut the events.

SIMULATION, HIGH-PASS STAGE ONLY



THE SIGNAL PATH

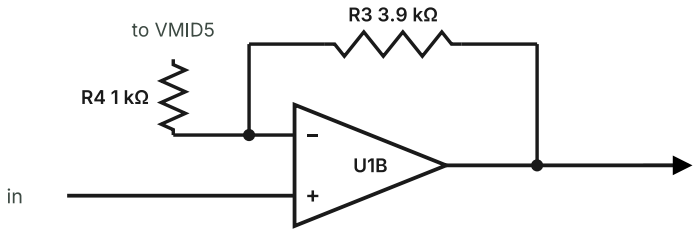
10 How the gain is set

The second half of U1 is a non-inverting amplifier. Its gain comes from two resistors: R3 from the output back to the minus input, and R4 from that input to VMID5.

$$G = 1 + \frac{R_3}{R_4} = 1 + \frac{3.9\,\text{k}\Omega}{1\,\text{k}\Omega} = 4.9$$

Because R4 goes to VMID5 and not to ground, the gain is 1 for DC. The stage output rests at 2.5 V, and the gain applies only to the burst.

GAIN 4.9



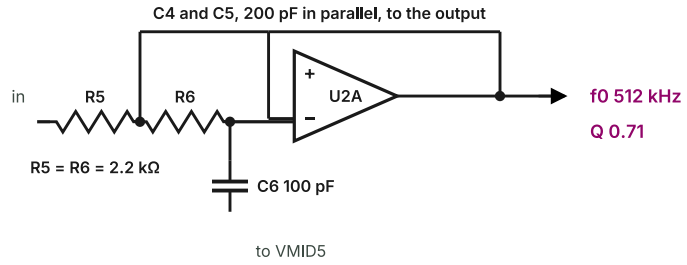
11 THE SIGNAL PATH
How the low-pass works

The low-pass is also a Sallen-Key stage, with the parts swapped: two resistors in series and two capacitors. The capacitors give high frequencies a path to VMID5.

$$f_0 = \frac{1}{2\pi R\sqrt{C_1 C_2}}, \qquad Q = \frac{1}{2}\sqrt{\frac{C_1}{C_2}}$$

With R = 2.2 kΩ, C1 = 200 pF (C4 and C5 in parallel) and C2 = 100 pF (C6), the corner is 512 kHz. Q is 0.71.

SALLEN-KEY LOW-PASS

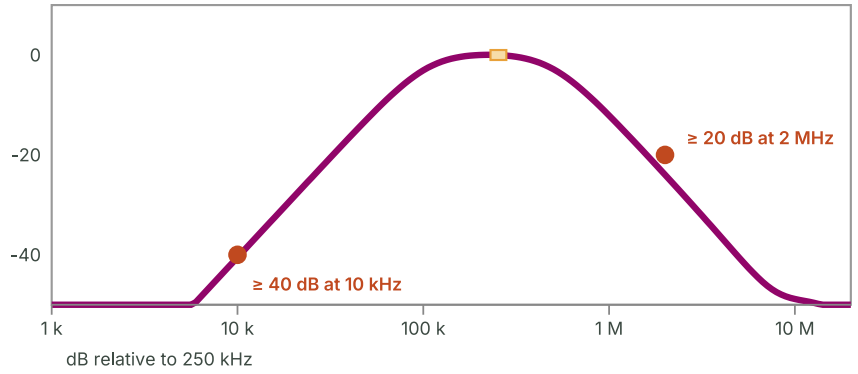


12 THE SIGNAL PATH
The total response

Put the three stages together and add the 50 Ω source. The simulation, with TI's own OPA365 model, gives 12.3 dB of gain at 250 kHz. That is ×4.1, a little below the ideal ×4.9.

The loss at 10 kHz is 40.5 dB against a target of 40 dB. With part tolerances 11 of 150 runs fall under 40 dB, the worst at 39.7 dB.

SIMULATION, SMA TO ADC

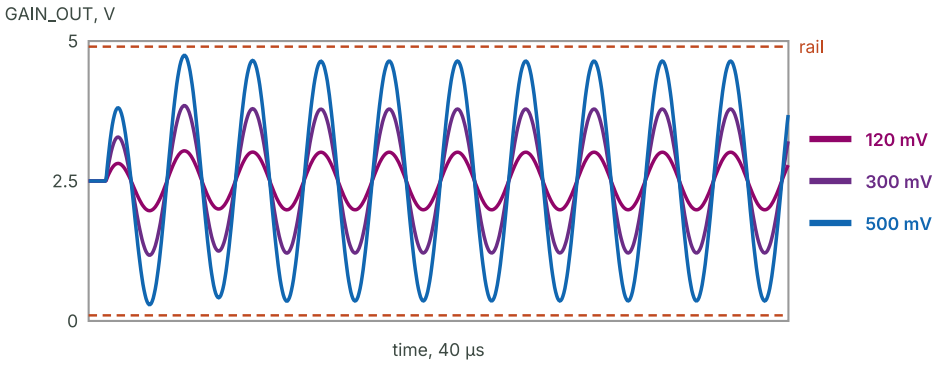


13 THE SIGNAL PATH
What clips first

The 5 V stages are the narrowest part of the path. They can swing about 2.4 V either side of 2.5 V. A burst clips there when it is larger than 0.54 V at the perfboard output.

A typical recorded event is 120 mV, so the margin is 4.5×. At the ADC that event fills 16 % of the full scale.

SIMULATION, 3 BURSTS

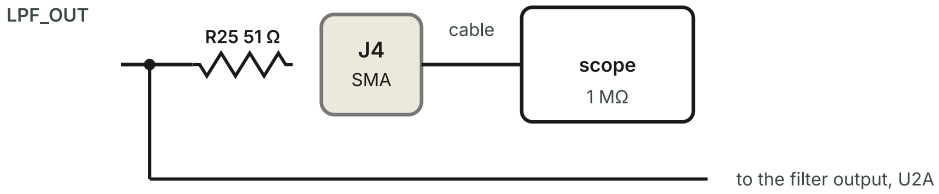


14 THE SIGNAL PATH
What SCOPE OUT is for

J4 gives the scope the same signal that the ADC sees, taken at the output of the low-pass. A 51 Ω resistor (R25) sits between the op-amp and the connector.

The resistor keeps the cable capacitance away from the op-amp, so the filter cannot ring. Use the 1 MΩ input of the scope. A 50 Ω input would form a divider with R25 and halve the signal.

CABLE LOAD



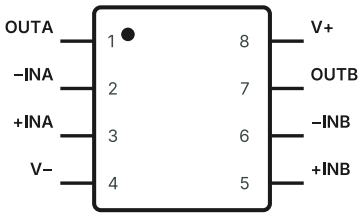
1 MΩ input: 1.000 of LPF_OUT. 50 Ω input: 0.50

15 THE SIGNAL PATH
What the OPA2365 does here

Two dual op-amps, U1 and U2, do all the active filtering. The OPA2365 is rail-to-rail on input and output and runs from 2.2 to 5.5 V. Its gain-bandwidth is 50 MHz and its noise is 4.5 nV/√Hz.

Each package holds two channels. The table shows which stage uses which channel.

SOIC-8, TOP VIEW



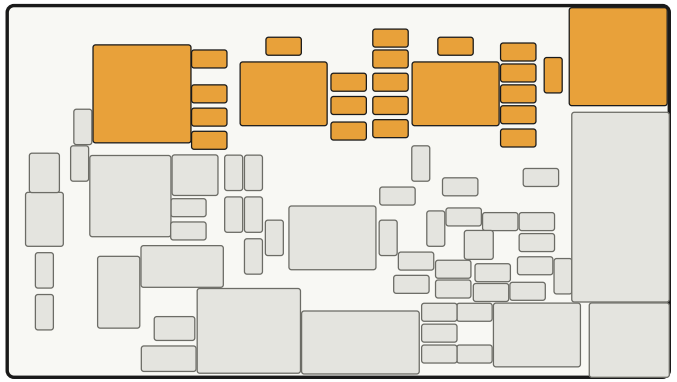
U1A pins 1 2 3	high-pass buffer
U1B pins 5 6 7	gain × 4.9
U2A pins 1 2 3	low-pass buffer
U2B pins 5 6 7	VMID5 buffer

16 THE SIGNAL PATH
Where these parts are

The signal parts sit in the top half of the board, from J1 on the left to U2 on the right. U1 holds the high-pass and the gain. U2 holds the low-pass and the VMID5 buffer.

J4 and its resistor R25 stand in the top right corner, away from the ADC.

TOP VIEW, ACTIVE PARTS IN AMBER



Input and high-pass
C1 C2 R1 R2

Gain
R3 R4

Low-pass
R5 R6 C4 C5 C6

Op-amps
U1 U2 C3 C10

VMID5
R7 R8 C7 R9 C8 C9

Connectors
J1 J4 R25

NOTES

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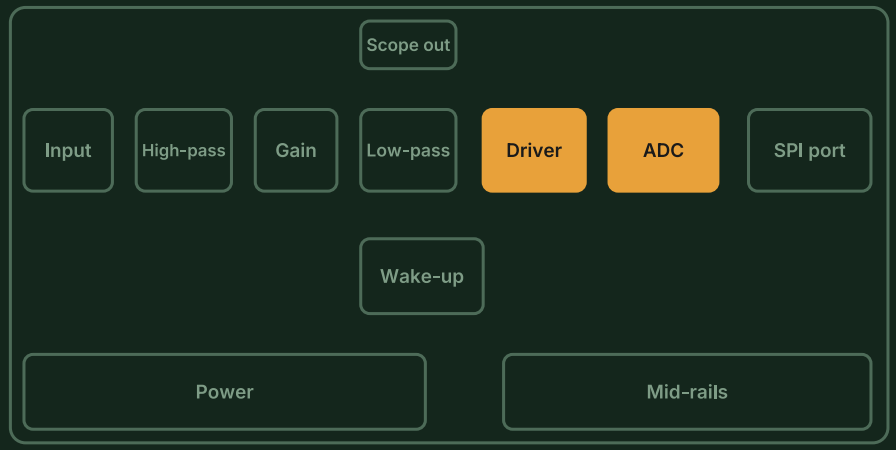
SECTION 3

Driver and ADC

STEPS 17 TO 24

How one wire becomes two, how the ADC input is protected from its own sampling kick, what the ADS7057 is and what its numbers mean for this board.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

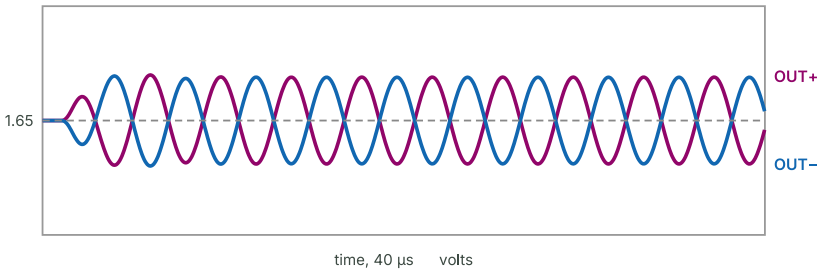
17 DRIVER AND ADC
How one wire becomes two

The ADS7057 needs two inputs that move in opposite directions. The **THS4531** (U3) makes them from one wire. The capacitor C11 and R10 bring the signal in. The corner of this coupling is 159 Hz, far below the band.

$$V_{OUT+} - V_{OUT-} = \frac{R_{12}}{R_{10}} V_{IN} = 1 \cdot V_{IN}$$

Each output moves half of the signal around VOCM, in opposite directions.

SIMULATION, 120 mV BURST



18 DRIVER AND ADC
What to know about the THS4531

Three facts from the datasheet shape the board. The **PD** pin must be driven, so U3 ties it to 5 V. The **VOCM** pin accepts 0.95 to 4.0 V, so 1.65 V is fine.

The third fact is a limit. The designers of the part expect about 2 pF of load between its outputs. A larger load needs a series resistor.

DATASHEET AGAINST THIS BOARD

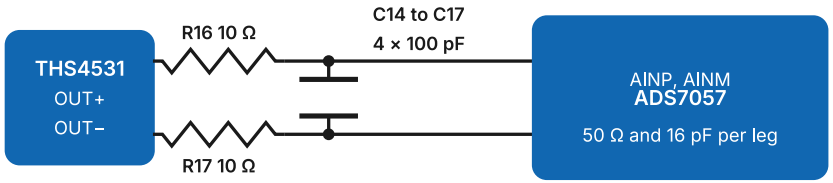
item	datasheet	this board	
Supply	2.5 to 5.5 V	**5 V**	
Gain-bandwidth	27 MHz	**gain 1**	
Noise	10 nV/√Hz	**0.2 LSB**	
VOCM range at 5 V	0.95 to 4.0 V	**1.65 V**	
PD pin	must be driven	**tied to 5 V**	
Load between outputs	2 pF nominal	**400 pF + 20 Ω**	OPEN

19 DRIVER AND ADC
What the 10 Ω and 400 pF are for

When the ADC samples, its 16 pF capacitor connects to the input and pulls a burst of charge. A step of 0.56 V moves 9 pC. A big capacitor next to the pins supplies this charge, so the driver sees only a small glitch.

Four 100 pF capacitors make 400 pF, which is 25 times the sampling capacitor. The datasheet asks for at least 20 times. The two 10 Ω resistors isolate the driver. The pole is at 20 MHz, far above the band.

KICKBACK FILTER



the node dips by 22 mV (53 LSB) and recovers in 54 ns at worst

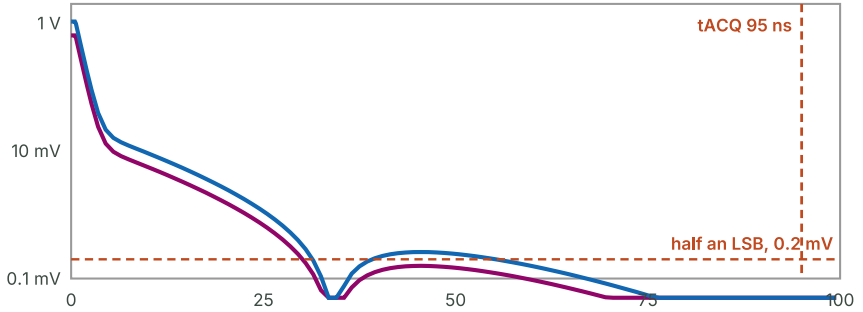
the 25× rule: 16 pF × 20 = 320 pF minimum

20 DRIVER AND ADC
What the ADC input looks like

The sampling switch of the ADC is 50 Ω in series with 16 pF on each leg. The simulation closes the switch on a signal that was left at the previous sample. The input settles to half an LSB in 29 ns for a typical step and 54 ns for a large one.

The ADC needs the input settled within 95 ns at the highest rate. So the margin is a factor of about two on the worst case.

SIMULATION, SETTLING



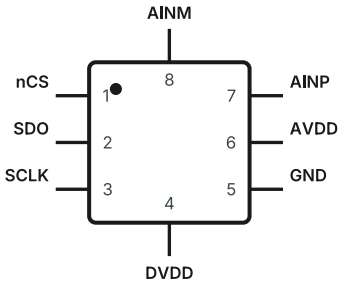
ns after the switch closes. Error against the final value, log scale.

21 DRIVER AND ADC
What the ADS7057 is

A 14-bit successive-approximation ADC in a 1.5 × 1.5 mm package with eight pads. It samples on the falling edge of CS and sends its result over SPI.

The table lists the numbers that matter for this board. All come from datasheet SBAS821.

X2QFN-8, TOP VIEW



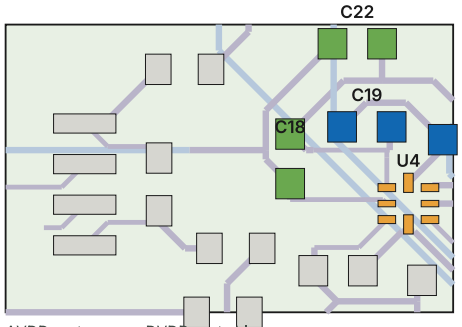
Resolution	14 bit, no missing codes
Rate	up to 2.5 MSPS
Supply	AVDD 2.35 to 3.6 V, DVDD 1.65 to 3.6 V
Input	differential, ±AVDD, common mode AVDD/2
SNR	79.6 dB at 2 kHz
INL, DNL	±0.9 LSB, ±0.3 LSB typical
SPI	mode 1, SCLK up to 60 MHz
Conversion	18 SCLK, then at least 95 ns acquisition

22 DRIVER AND ADC
How the ADC is decoupled

AVDD gets a 4.7 µF (C18) and a 100 nF (C22). DVDD gets a 1 µF (C19) and a 100 nF (C20). The datasheet asks for 3.3 µF and 1 µF and says to keep them close, with no vias between the pin and the capacitor.

On this board C22 is 4.4 mm from the ADC and C18 is 3.0 mm. C20 is 1.6 mm and C19 is 1.8 mm. The 100 nF parts sit farther than the large ones, which is a point to improve.

REAL BOARD, 7 ×



AVDD parts green, DVDD parts blue

23 DRIVER AND ADC
How a voltage becomes a code

The 120 mV event of the simulation reaches 0.517 V between AINP and AINM. The code is the voltage divided by 1 LSB, rounded down.

$$\text{code} = \left\lfloor \frac{V_{\text{AINP}} - V_{\text{AINM}}}{1\ \text{LSB}} \right\rfloor$$

So the peak is 1284, which is 0x0504. The negative peak -0.507 V gives -1258, which is 0x3B16 in two's complement.

FROM THE SIMULATION

D13 is the sign bit

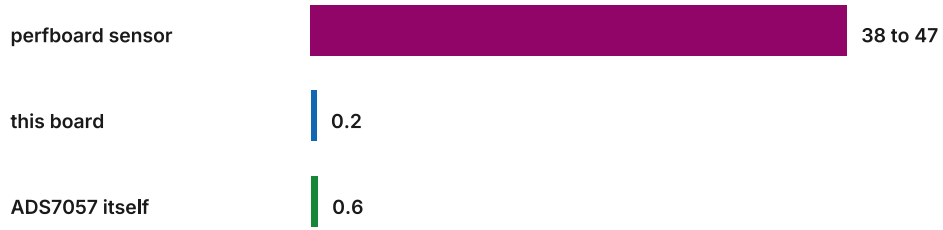
	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	
positive peak	0	0	0	1	0	1	0	0	0	0	0	1	0	0	0x0504
negative peak	1	1	1	0	1	1	0	0	0	1	0	1	1	0	0x3B16

24 DRIVER AND ADC
How much noise the ADC sees

Three noise sources add. The sensor noise of the perfboard, through this board, is 38 to 47 LSB rms. The board adds 0.2 LSB. The ADC itself adds about 0.6 LSB.

So the sensor decides the noise floor. The board and the ADC are small by comparison, and this is what a test board should show.

LSB RMS, ESTIMATE FOR THE ADC

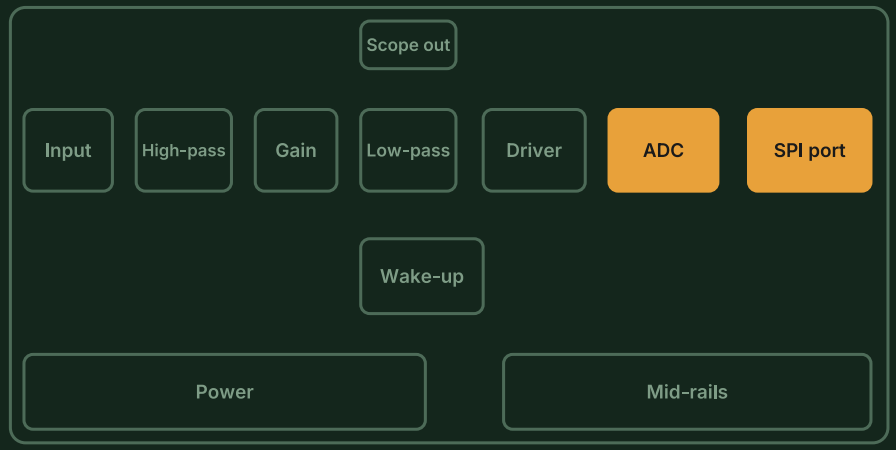


SECTION 4

The SPI link

STEPS 25 TO 30

The four wires from the ADC to the ZedBoard, what one frame looks like, why the data comes one frame late, the calibration at power-up and how fast the link can go.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

25 THE SPI LINK
Which pin is which on the PMOD

The board plugs into one PMOD port of the ZedBoard, JA to JD. The board uses eight of the twelve pins. **CS**, **SCLK** and **SDO** carry SPI. **nWAKE** is the wake-up line. 3V3 powers the ADC logic, and GND is the return.

There is no MOSI wire, because this ADC only sends data. The numbering follows the Digilent SPI PMOD, so pin 1 is CS.

J3, SEEN FROM THE TOP

left column	right column
1 CS	7 not used
2 not used	8 not used
3 SDO	9 not used
4 SCLK	10 nWAKE
5 GND	11 GND
6 3V3	12 3V3

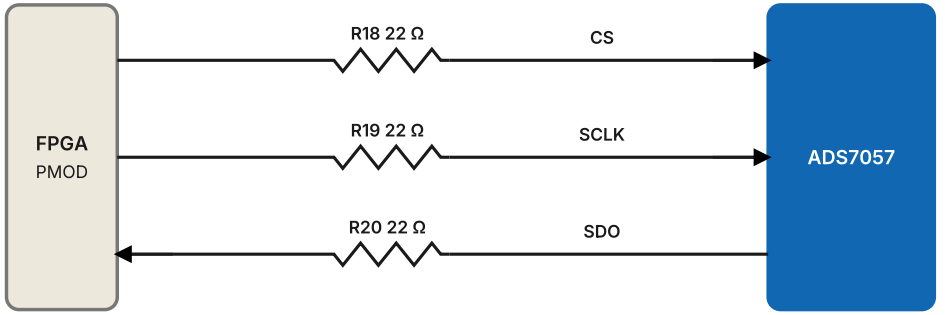
The number is the PMOD pin. Row 1 holds pin 1 and pin 7.

26 THE SPI LINK
Why 22 Ω in each line

R18, R19 and R20 sit in the CS, SCLK and SDO lines at the ADC. Together with the small capacitance of the trace and the pin they slow the edge of the FPGA output a little and damp ringing.

The cost in speed is tiny. 22 Ω with about 10 pF (ESTIMATE) gives 0.2 ns, and one bit at 60 MHz lasts 16.7 ns.

SERIES RESISTOR

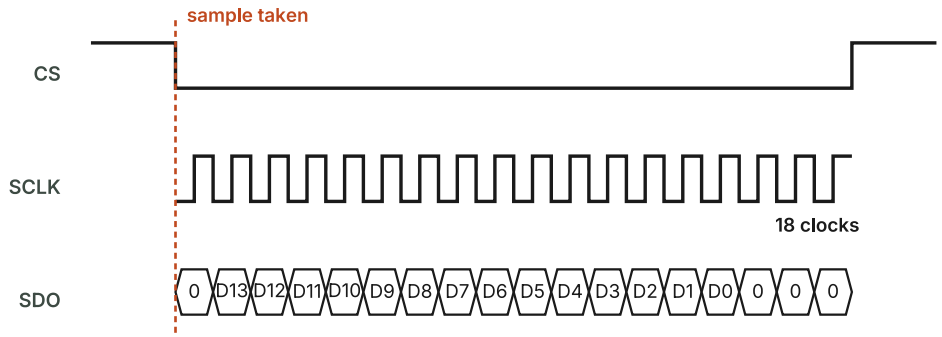


27 THE SPI LINK
What one frame looks like

A **frame** starts when CS falls. The ADC samples at that moment. The host then sends 18 clock pulses. On SDO the ADC answers with a leading 0, then 14 data bits, most significant bit first, then three zeros.

After the 18th clock the ADC converts nothing more. CS goes high and the ADC acquires the next sample for at least 95 ns.

ONE FRAME

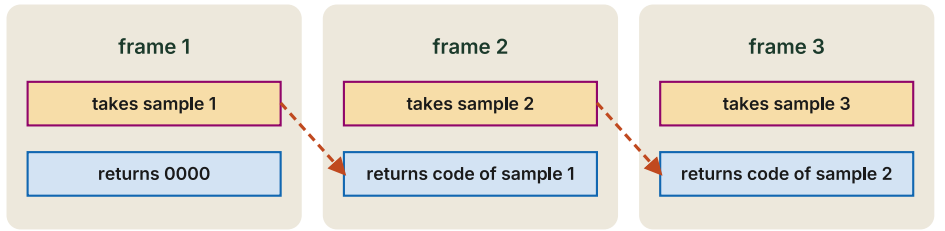


28 THE SPI LINK
Why the data is one frame late

The ADC uses the clocks of a frame to send the result of the **previous** conversion. The conversion of the current sample runs inside the same frame, and its result waits for the next one.

So the first frame after power-up returns all zeros, and every later frame returns the sample that was taken one frame before.

THREE FRAMES

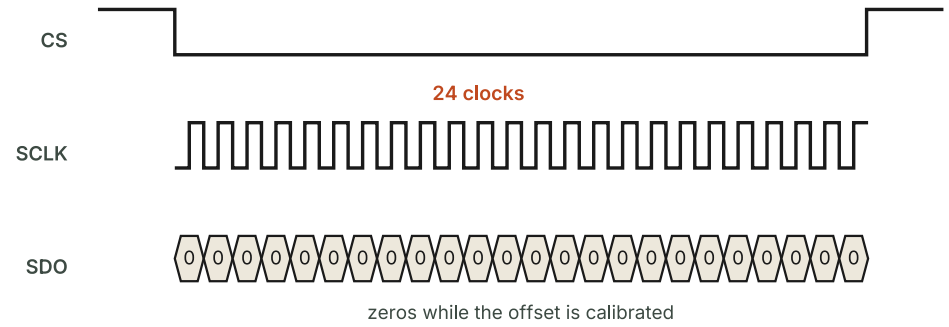


29 THE SPI LINK
What happens at power-up

The datasheet recommends an offset calibration at power-up. The host does it by sending **24** clocks in the first frame, not 18. The ADC then disconnects its input, corrects its own offset and sends zeros.

At a 20 MHz clock this first frame takes 1200 ns plus 95 ns of acquisition, so 1295 ns. Repeat it when AVDD or the temperature changes a lot.

FIRST FRAME



30 THE SPI LINK
How fast the link can go

One sample costs 18 clock periods plus 95 ns. So the sample rate is set by SCLK. The datasheet allows up to 60 MHz, which gives the full 2.5 MSPS.

There are no level shifters on this board, so the FPGA drives the ADC pins directly at 3.3 V. Start at 10 to 20 MHz and raise the clock when the first test works.

SAMPLE RATE AGAINST SCLK

SCLK	one sample	sample rate
10 MHz	1895 ns	**0.53 MSPS**
20 MHz	995 ns	**1.01 MSPS**
25 MHz	815 ns	**1.23 MSPS**
40 MHz	545 ns	**1.83 MSPS**
60 MHz	395 ns	**2.50 MSPS**

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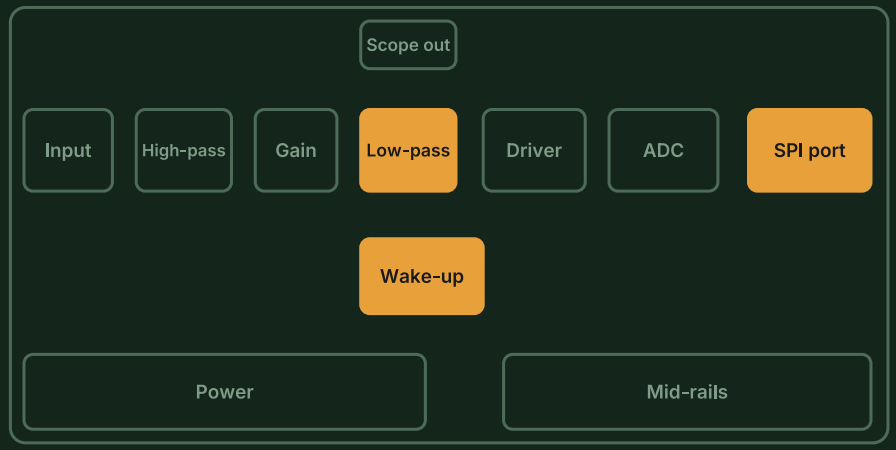
SECTION 5

The wake-up

STEPS 31 TO 35

How the LM393 turns a strong burst into a low level on nWAKE, how the trimmer sets the window and how to set it on the bench.



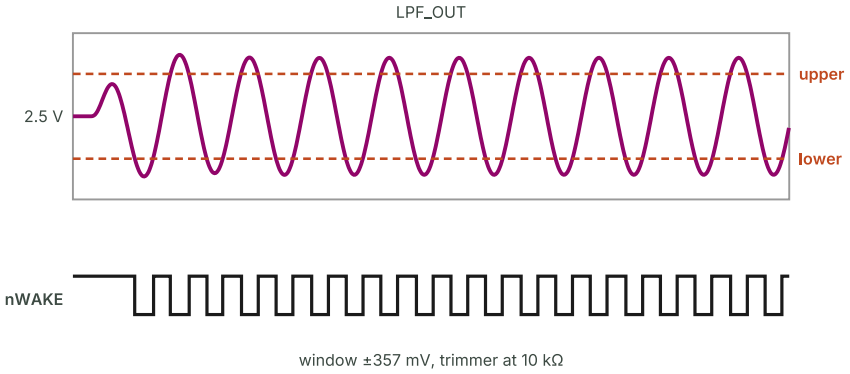
BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

31 THE WAKE-UP
What a window comparator is

The LM393 holds two comparators. One trips when the signal rises above the **upper** threshold. The other trips when it falls below the **lower** threshold. Both thresholds sit around 2.5 V.

The two outputs share one wire. So nWAKE goes low as soon as the signal leaves the window on either side.

SIMULATION, 120 mV BURST



window ±357 mV, trimmer at 10 kΩ

32 THE WAKE-UP
How the window is set

One resistor chain from 5 V to ground makes both thresholds. The trimmer RV1 and a 1 kΩ resistor set the distance between them.

$$V_{\text{half}} = 2.5\,\text{V} \cdot \frac{1\,\text{k}\Omega + R_p}{67\,\text{k}\Omega + R_p}$$

The last column converts the window back to the perfboard output with a gain of 4.1.

RP AGAINST WINDOW

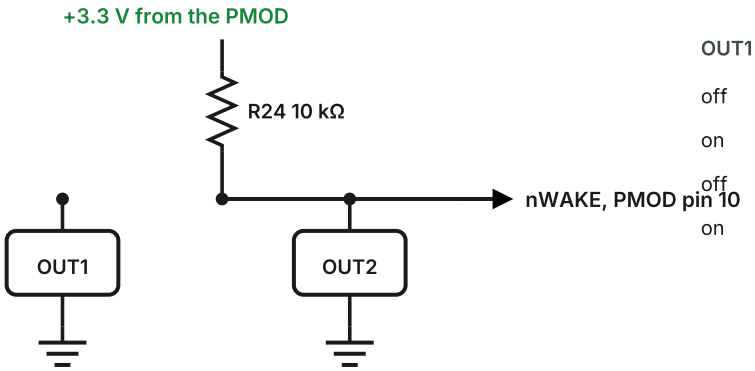
Rp	upper	lower	half window	at J1
0 kΩ	2.537 V	2.463 V	37 mV	9 mV
5 kΩ	2.708 V	2.292 V	208 mV	51 mV
10 kΩ	2.857 V	2.143 V	357 mV	87 mV
14 kΩ	2.963 V	2.037 V	463 mV	113 mV
20 kΩ	3.103 V	1.897 V	603 mV	147 mV

33 THE WAKE-UP
Why nWAKE is open-collector

The LM393 output can only pull the line to ground. It cannot drive it high. A 10 kΩ resistor (R24) pulls the line up to 3.3 V from the PMOD.

So the FPGA never sees 5 V, although the LM393 runs from 5 V. The two outputs share one wire, so either one can pull nWAKE low.

WIRED-OR



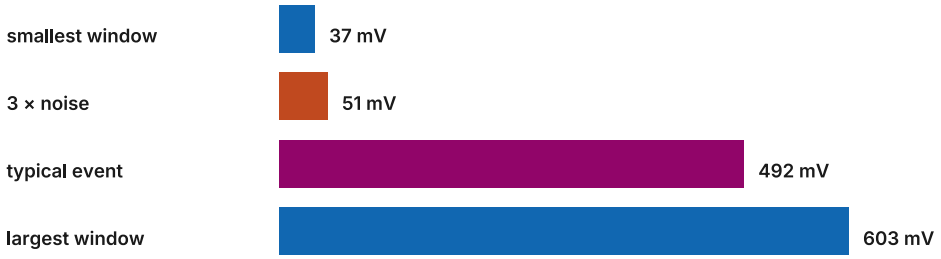
OUT1	OUT2	nWAKE
off	off	high
on	off	low
off	on	low
on	on	low

34 THE WAKE-UP
How fast and how accurate

The LM393 answers in about 1.3 µs. That is slow for a clock but fast for a wake-up. Its input offset is about 5 mV, which is small against the window.

The noise decides the smallest useful setting. The quiet baseline of 4.1 mV rms becomes 17 mV rms at the comparator. Three times that is 51 mV. Below Rp = 0.4 kΩ the noise alone trips the output.

NOISE AGAINST WINDOW, ESTIMATE



THE WAKE-UP

35 How to set the trimmer

Choose the burst that should wake the system. Feed that burst to J1. Watch SCOPE OUT and nWAKE together. Turn RV1 slowly until nWAKE drops for that burst and stays high without it.

A 50 mV burst at J1 needs Rp near 4.9 kΩ. A 100 mV burst needs Rp near 12.0 kΩ. The trimmer is a 20 kΩ part, so the middle of its travel is close to 100 mV.

BENCH STEPS

1. **Feed 250 kHz, 100 mV to J1**
2. **Read 0.41 V peak on SCOPE OUT**
3. **Turn RV1 until nWAKE just pulses**
4. **Remove the burst. nWAKE must stay high**

clockwise should widen the window. Check with a meter on VTH_H

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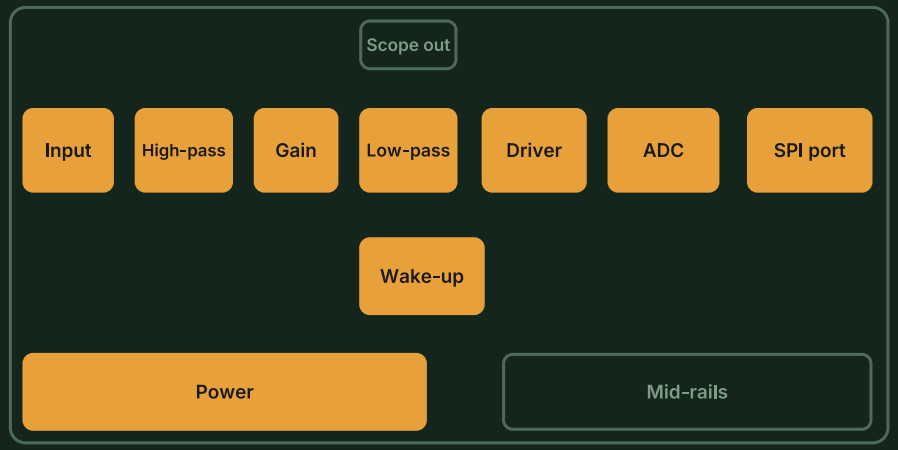
SECTION 6

The board

STEPS 36 TO 44

The stack, the floor plan, where every part sits, how the tracks and the power are routed, what the silkscreen says and which footprints need care.



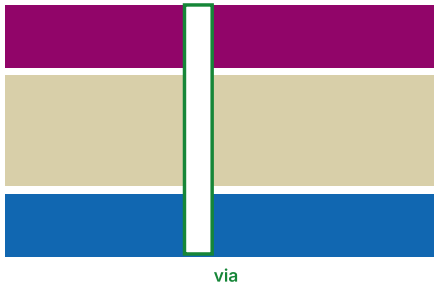
BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

36 THE BOARD
The stack and the size

The main board is 57 × 32 mm with two copper layers on 1.6 mm FR-4. Both layers carry a ground pour. 122 of the 148 vias are ground stitching vias that tie the two pours together.

All 475 track segments are 0.2 mm wide, except 67 short segments of 0.15 mm near the fine-pitch pads. Vias are 0.6 mm with a 0.3 mm drill.

FROM THE BOARD FILE



top copper 431 track segments, all parts, ground pour

FR-4, 1.6 mm fibreglass epoxy

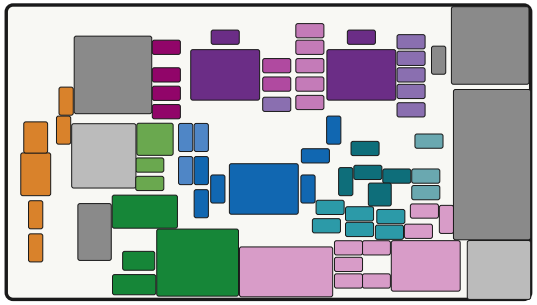
bottom copper 44 track segments, ground pour

37 THE BOARD
How the floor is planned

The signal runs left to right across the top half, as in the block diagram. The driver and the ADC sit in the middle. The power parts take the bottom left. The comparator sits in the bottom right.

The PMOD connector is on the right edge so the board hangs on the ZedBoard. The SMA connectors stand at the top, away from the ADC. The left strip holds the bias-T.

TOP VIEW, COLORS BY BLOCK



- Input and high-pass
- Gain
- Low-pass
- Op-amps U1, U2 and bypass
- Mid-rail VMID5
- Driver THS4531
- VOCM 1.65 V
- ADC input filter
- ADS7057 and bypass
- SPI series resistors
- AVDD regulator
- 9 V to 5 V
- Wake-up comparator
- Bias-T for the head
- Connectors

38 THE BOARD
Where every part is

The table lists the blocks, the references and the count. The main board has 75 footprints: 66 surface-mount parts, 5 through-hole parts and 2 mounting holes, plus one not-fitted capacitor and the solder jumper JP1.

Every part has a place in exactly one block, and the same reference appears in the schematic, the BOM and the placement file.

BLOCKS, REFERENCES

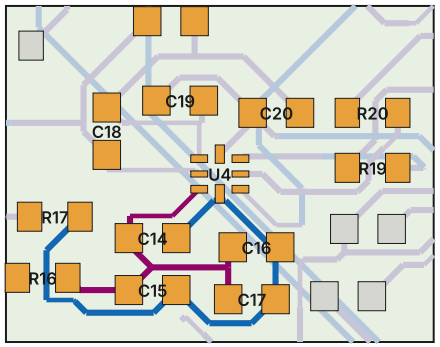
- **Input and high-pass** — C1 C2 R1 R2
- **Gain** — R3 R4
- **Low-pass** — C4 C5 C6 R5 R6
- **Op-amps U1, U2 and bypass** — C3 C10 U1 U2
- **Mid-rail VMID5** — C7 C8 C9 R7 R8 R9
- **Driver THS4531** — C11 C12 R10 R11 R12 R13 U3
- **VOCM 1.65 V** — C13 R14 R15
- **ADC input filter** — C14 C15 C16 C17 R16 R17
- **ADS7057 and bypass** — C18 C19 C20 C22 U4
- **SPI series resistors** — R18 R19 R20
- **AVDD regulator** — C21 C30 U5
- **9 V to 5 V** — C23 C29 D1 U7
- **Wake-up comparator** — C26 C27 C28 C31 RV1 R21 R22 R23 R24 U6
- **Bias-T for the head** — C32 C33 JP1 L2 R26 R27
- **Connectors** — J1 J2 J3 J4 R25
- **Holes** — H1 H2

39 THE BOARD
Why the ADC corner is built this way

Everything that touches the ADC input sits around U4: the two 10 Ω resistors, the four 100 pF capacitors, the supply capacitors and the three 22 Ω resistors. The tracks AINP and AINM are 12 and 18 mm long.

The 5 mm difference is harmless here. A 5 mm track delays a signal by about 30 ps, and a 500 kHz wave lasts 2 µs.

REAL BOARD, ADC CORNER

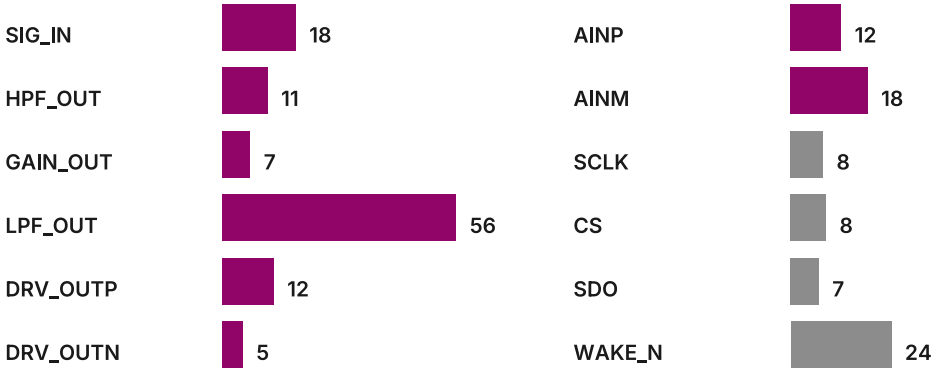


40 THE BOARD
How the analog tracks are routed

The longest analog net is LPF_OUT at 56 mm, because it feeds the SCOPE OUT connector, the comparator and the driver. All other analog nets are under 24 mm.

The router chose the paths, and the board is small enough that none of them matters at 500 kHz. The figure lists the lengths.

NET LENGTH, mm

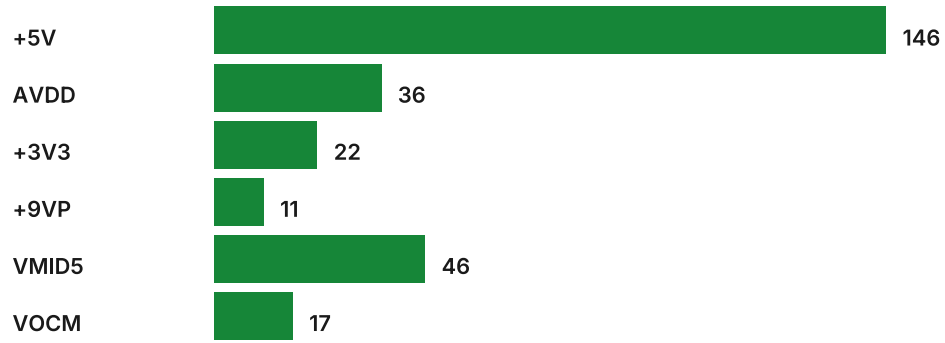


41 THE BOARD
How power and ground are routed

Ground is a pour on both layers, tied by 122 vias, so every ground pad finds a short path. The power nets are ordinary 0.2 mm tracks, not wide ones, because the currents are small.

The +5 V net is 146 mm of track in total. With 27 mA (22 mA, plus 5 mA for the sensor head) this drops about 10 mV across its whole length (ESTIMATE, worst case).

POWER NET LENGTH, mm

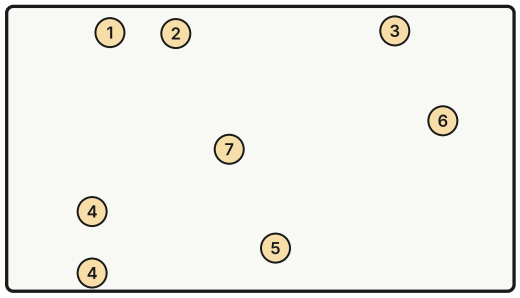


42 THE BOARD
What each label on the silkscreen means

The front carries the labels that a user needs. The names of the parts (R16, C29 and so on) are not on the silkscreen, because the board is dense. They are in the KiCad file and in the 1:1 print.

The back carries the PMOD pin list and the title.

FRONT SIDE



1 SENSOR IN: J1, sensor head, +5 V on the centre
2 CH box: write the channel number
3 SCOPE OUT: J4, the signal at the ADC
4 +9V and GND: battery, square pad is +
5 WAKE LEVEL: trimmer RV1
6 1: PMOD pin 1
7 logo: Auselia mark

43 THE BOARD
Which footprints are tricky

The ADS7057 pads are 0.55 × 0.25 mm at 0.5 mm pitch. The gap between pads is 0.25 mm, so the solder mask web is thin. JLCPCB assembled it in the quote without a warning.

Four other parts needed care. The factory model of each one had a different orientation or origin from the footprint, and the CPL now corrects it.

PARTS TO WATCH

U4 ADS7057
X2QFN-8, 1.5 mm
pin 1 mark checked

U5 XC6206
SOT-23, 3 pins
rotation 180

C29 tantalum
polarity: + on pad 1
rotation 0

J3 PMOD
right-angle 2 × 6
5.9 mm beyond the edge

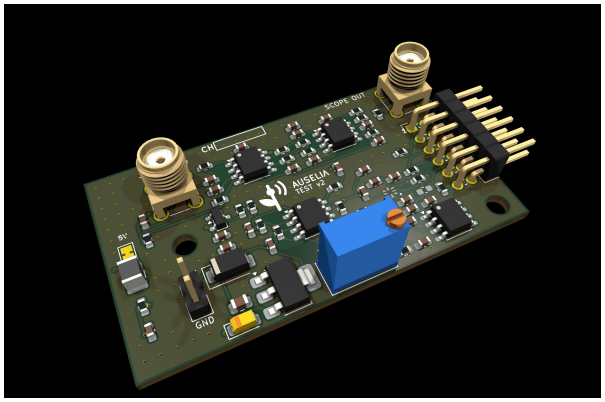
THE BOARD

44 What the board looks like

The render is from the final KiCad file. J1 stands on the top left and J4 on the top right. The PMOD header J3 points off the right edge. The blue trimmer RV1 is at the bottom.

The two holes are M3 mounting holes.

3D RENDER, KICAD



J1: SENSOR IN, top left

J4: SCOPE OUT, top right

J3: PMOD, right edge

RV1: wake level trimmer

U4: ADS7057, centre right

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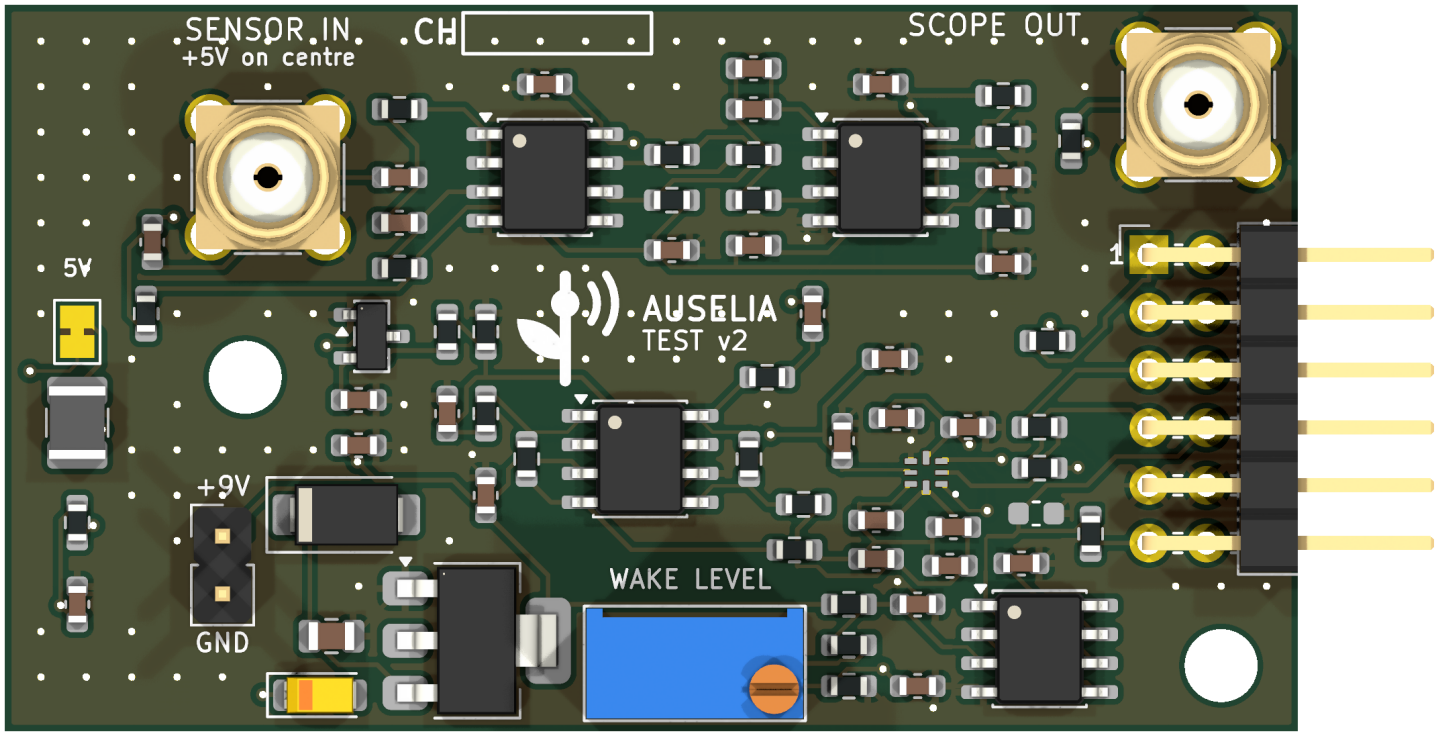
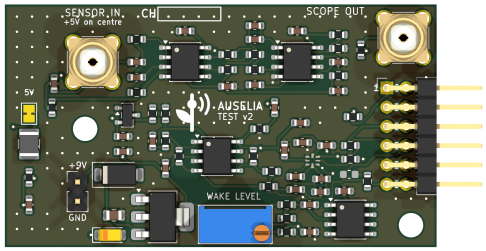


10 mm at this scale (3 : 1)

Board 57.0 × 32.0 mm, KiCad 3D render, view from the top



The same view at real size, 1 : 1

SECTION 7

Making the board

STEPS 45 TO 48

The files the factory needs, the parts list, what the placement viewer corrected and what the order costs.

MAKING THE BOARD

45 Which files go to the factory

JLCPCB needs three files. The Gerber zip makes the bare board. The BOM names every part. The CPL gives the position and angle of each part.

The main board zip holds 13 files. The BOM has 30 lines and 71 parts. The CPL has 71 rows. The mounting holes and the not-fitted capacitor C31 are in none of them. The sensor head has its own zip, BOM and CPL.

FILES IN hw/fab

ADC_Test_gerbers.zip
Copper, mask, silk, paste, outline, drill

ADC_Test_BOM_JLC.csv
Value, references, footprint, LCSC number

ADC_Test_CPL_JLC.csv
Reference, X, Y, layer, rotation

MAKING THE BOARD

46 What is in the BOM

30 lines hold 71 parts. Five lines hold extended parts, and JLCPCB charges a fee for each type. All other lines hold basic parts.

The ADC, the driver and the two op-amps are the extended parts, with the tantalum capacitor and the 470 µH inductor of the bias-T.

EXTENDED PARTS, PRICE PER PART

Ref	Part	LCSC	Price
U1,U2	OPA2365	C130011	$1.38
U3	THS4531	C705347	$2.99
U4	ADS7057	C2669902	$7.98
C29	22uF 10V tantalum	C11366	$0.19
L2	470uH	C75644	$0.09

Through-hole parts: J1 J4 SMA, J2, J3, RV1. JLCPCB solders them by hand, for a fee per joint.

MAKING THE BOARD

47 What the placement viewer corrected

The JLCPCB viewer showed four parts turned wrong on the first revision. The new parts (U8, L1, L2, J6) are not checked yet. The KiCad footprints and the JLC library disagree on the zero angle. Each fix sits in `jlc_files.py`, not in the board.

The trimmer RV1 also moved 2.54 mm in X and 1 mm in Y, because its library origin differs from the pin 1 pad.

KICAD ROTATION AGAINST CPL ROTATION

Ref	KiCad	CPL	Why
U5	0°	180°	viewer showed it turned; set absolute angle
C29	0°	0°	library offset would turn it; kept KiCad angle
J2	0°	90°	viewer showed it turned; set absolute angle
J3	0°	90°	viewer showed it turned; set absolute angle

Rule: look at every part in the viewer once. The viewer shows the real answer.

MAKING THE BOARD

48 What the order costs

Two ADC boards and five sensor heads are two JLCPCB orders. They cost $107.75 at the factory. With one parcel, shipping and IVA, the total is $141.01, or 135 thousand CLP at 960 CLP per USD.

The ADC order is $66.99 and the head order is $40.76. Five ADC boards would cost $118.20. These are estimates from the model of the first quote, until you upload the files.

ESTIMATE, 2 ADC BOARDS + 5 HEADS, USD

PCB $8.00 7.7 k CLP
Parts $47.96 46.0 k CLP
Fees $51.78 49.7 k CLP
Shipping $10.75 10.3 k CLP
IVA 19 % $22.51 21.6 k CLP

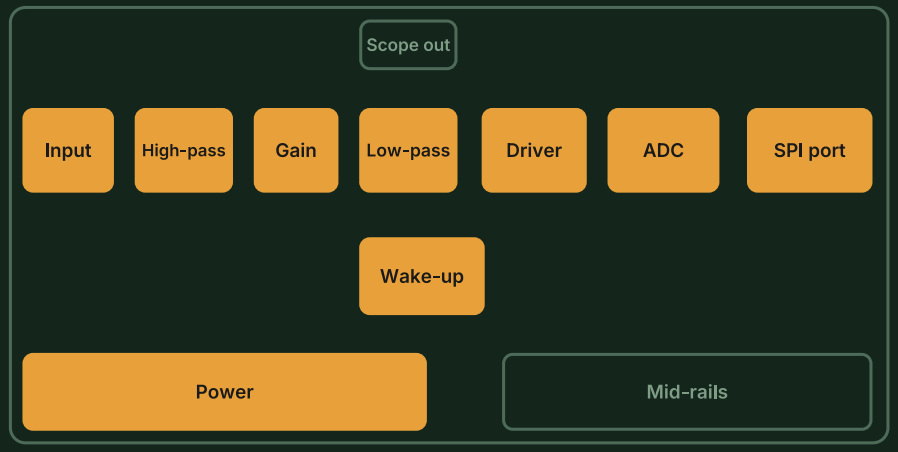
Total $141.01. Two parcels $153.81. 5 + 5 boards $201.95.

SECTION 8

Bring-up and checks

STEPS 49 TO 54

What to measure before the PMOD goes in, the first signal test, the response to expect, how to test the wake-up, what to do when something fails and what is still open.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

BRING-UP AND CHECKS

49 What to check before the PMOD goes in

Feed 9 V to J2 with the PMOD unplugged. Measure each rail against GND with a meter. Unplug the supply if the 5 V rail is off by more than 0.25 V.

The 3.3 V line comes from the ZedBoard through the PMOD. With no PMOD it reads 0 V, and that is correct.

EXPECTED VOLTAGES, 9 V IN

Node	Where	Expect
+5V	C29 plus side	5.0 V
AVDD	C18 or U4 pin 6	3.30 V
VMID5	U2 pin 7	2.50 V
VOCM	R14 and R15 joint	1.65 V
+3V3	J3 pin 11	0 V (no PMOD)
nWAKE	J3 pin 8	floats, no pull-up

BRING-UP AND CHECKS

50 The first signal test

Plug the PMOD in. Put 100 mV peak at 250 kHz on J1 from the Analog Discovery 2 and send a burst of ten cycles. Look at SCOPE OUT and read the SPI words.

The scope shows 410 mV peak above 2.5 V. The ADC sees 410 mV peak between its inputs, which is 1018 codes.

EXPECTED, 100 mV PEAK AT J1

Where	Expect
J1	100 mV peak, 250 kHz
SCOPE OUT (J4)	410 mV peak on 2.5 V
ADC input, differential	410 mV peak
Code, peak	about 1018 (0x03FA)
Code, no signal	near 0 (0x0000), ± 1

BRING-UP AND CHECKS

51 What response to measure

Sweep J1 at 100 mV and read the code amplitude. Divide by the amplitude at 250 kHz. The numbers below come from the simulation.

Component spread moves the 10 kHz point. In 11 of 150 random runs the loss at 10 kHz was below 40 dB.

SIMULATION, NOMINAL PARTS

Frequency	Gain SMA to ADC	Against 250 kHz
10 kHz	-28.2 dB	-40.5 dB
50 kHz	-0.5 dB	-12.7 dB
100 kHz	+9.2 dB	-3.1 dB
250 kHz	+12.3 dB	+0.0 dB
500 kHz	+9.4 dB	-2.8 dB
1000 kHz	+0.1 dB	-12.1 dB
2000 kHz	-11.7 dB	-24.0 dB

BRING-UP AND CHECKS

52 How to test the wake-up

Sweep RV1 and keep a burst at J1. nWAKE must drop for a burst above the threshold and stay high below it. The table gives the threshold for five trimmer positions.

The lowest setting is ± 9 mV at J1. The quiet perfboard noise is 4.6 mV rms in the band, so that setting is only about 2 times the noise.

COMPUTED FROM THE RESISTORS

Trimmer	RV1	Threshold at J1
0 %	0.0 Ω	± 9 mV
25 %	5 kΩ	± 51 mV
50 %	10 kΩ	± 87 mV
75 %	15 kΩ	± 119 mV
100 %	20 kΩ	± 147 mV

ESTIMATE: comparator offset (up to 5 mV, datasheet) moves each threshold by that much.

BRING-UP AND CHECKS

53 What if it does not work

Find the first stage that is wrong, then look only at that stage. Work from the input toward the ADC. Measure DC first, then the signal.

Most first-board failures are a short, a part turned the wrong way or a missing 3.3 V.

SYMPTOM, CAUSE, CHECK

Symptom	Likely cause	Check
5 V low	Battery under 6.7 V	J2 voltage
SCOPE OUT flat	U1 or U2 not powered	5 V at pin 8
SDO all zeros	First frame only	Read twice
SDO stuck	No 3.3 V from PMOD	J3 pin 11
Wake always low	RV1 near 0 and noise	Turn RV1 up
Code offset	VOCM or VMID off	Step 49 table

NOTES

BRING-UP AND CHECKS

54 What is verified and what is open

Checked by tools: the electrical rules, the design rules, the netlist against the board, and the rotations in the JLC viewer. Checked by simulation: gain, band, noise and settling.

Nothing has run on a real board yet. Red rows mark the three open items.

STATUS

Item	Status	How
ERC, DRC, parity	Done	KiCad, 0 errors
Rotations	Done	JLC viewer
Chain response, noise	Simulated	ngspice, vendor models
Head and bias-T	Simulated	ngspice, OPA365
THS4531, 400 pF load	OPEN	LTspice netlist, not run
PMOD fit on ZedBoard	OPEN	print 1:1 and test
First board, bench	OPEN	steps 49 to 52

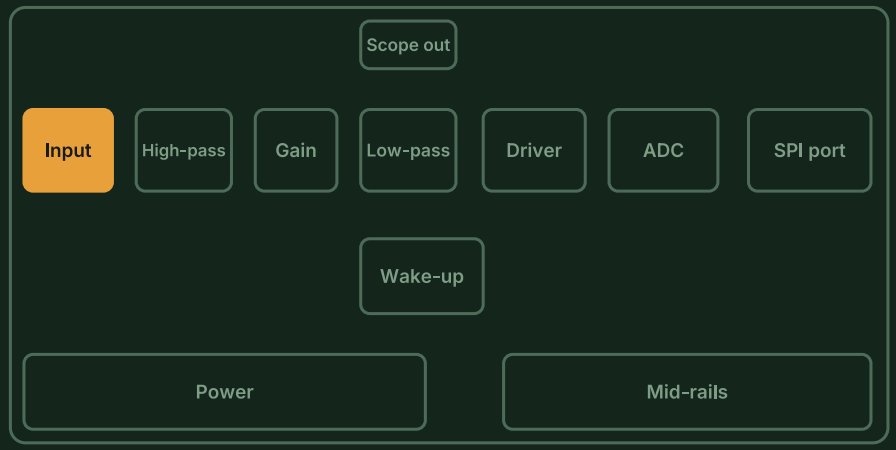
NOTES

SECTION 9

The sensor head

STEPS 55 TO 60

The perfboard charge amplifier on its own small board, how it is powered through the coax by a bias-T, how it is laid out and how to test it.



BLOCKS AT WORK IN THIS SECTION (SEE PAGE 3)

55 THE SENSOR HEAD
What the sensor head does

The head is the perfboard on a PCB. The piezo wires solder to it. It turns the charge of the piezo into a voltage and sends that voltage down one coax cable.

The same coax brings 5 V from the ADC board to the head. So the head needs one cable and no battery.

CHAIN

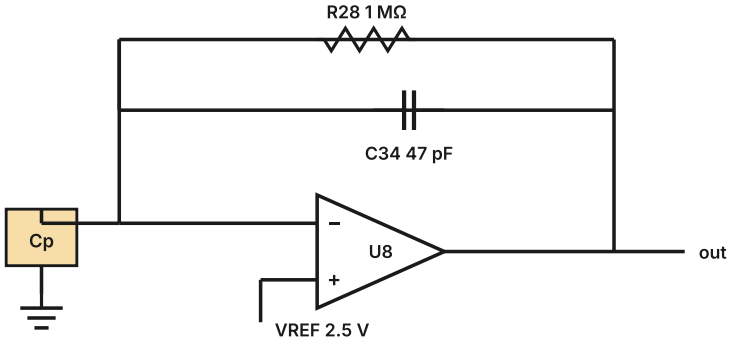
The head loses 0.16 dB at 250 kHz to the bias-T. To use a generator or a perfboard instead, cut the solder jumper JP1.

56 THE SENSOR HEAD
How the charge becomes a voltage

The amplifier keeps its input at a fixed voltage. So the charge Q of the piezo flows into the feedback capacitor Cf, and the output moves by Q / Cf. With Cf = 47 pF, a charge of 4 to 7 pC gives 85 to 149 mV.

Rf = 1 MΩ drains the capacitor with a time constant of 47 µs, a corner at 3.4 kHz. The noise gain is 35, so the 50 MHz amplifier keeps 1.4 MHz of bandwidth.

FROM THE NETLIST, ESTIMATE FOR Q

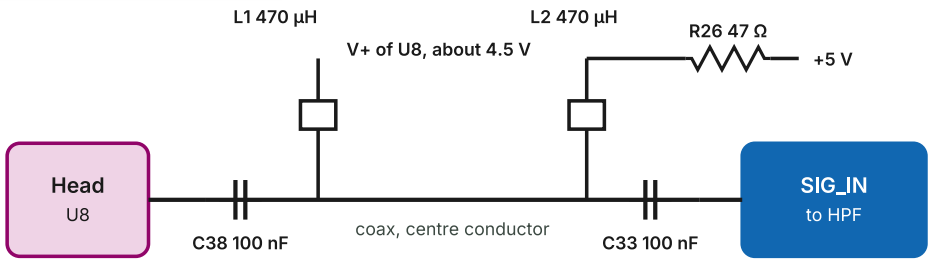


57 THE SENSOR HEAD
How the coax carries power and signal

A bias-T joins a DC path and an AC path on one wire. The DC path is an inductor: 470 µH looks like 738 Ω at 250 kHz, so it hardly loads the signal. The AC path is a capacitor: 100 nF passes 250 kHz and blocks the DC.

The simulation with the OPA365 model gives a loss of 0.16 dB at 250 kHz and 19 dB at 10 kHz. The low-frequency noise of the perfboard is cut before it reaches the filters.

SIMULATION, RG174 1 M



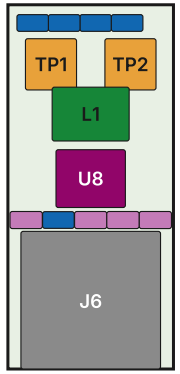
L1 and L2 are the same part. Cut the solder jumper JP1 (between C32 and L2) to turn the 5 V off.

58 THE SENSOR HEAD
Where the parts are on the head

The head is 10 × 22 mm, with 14 parts, all on one face. The piezo sits on the other face, flat against the pads. The SMA stands on the parts face and points out of the capsule.

The capsule drawing v3 has a PCB of 10 × 20 mm. With the SMA and the inductor the head needs 2 mm more, so the PCB sticks out 9 mm instead of 7.

REAL BOARD, TOP VIEW, 2.2 : 1



U8	OPA365, charge amplifier
L1	470 µH, supply from the coax
TP1	piezo +, summing node
TP2	piezo −, ground
J6	SMA, vertical
R28 C34	Rf 1 MΩ, Cf 47 pF
R29 R30 C35	VREF 2.5 V
C36 C37	supply bypass
R31 C38	49.9 Ω and 100 nF out

THE SENSOR HEAD

59 What goes to the factory for the head

The head has its own Gerber zip, BOM and CPL. It is 9 BOM lines and 12 parts. The OPA365 and the inductor come from the extended library.

Five heads cost about $40.76 at the factory, $8.15 each (ESTIMATE). Fixed fees are most of it: setup, stencil and hand soldering of the SMA.

HEAD BOM, PRICE PER PART

Ref	Part	LCSC	Price
C34	47pF C0G	C1567	$0.008
C35,C37	10uF	C15525	$0.026
C36,C38	100nF	C307331	$0.010
J6	SMA jack	C496549	$0.402
L1	470uH	C75644	$0.090
R28	1M	C26083	$0.002
R29,R30	10k	C25744	$0.002
R31	49.9R	C25120	$0.002
U8	OPA365	C96626	$2.674

NOTES

THE SENSOR HEAD

60 How to test the head

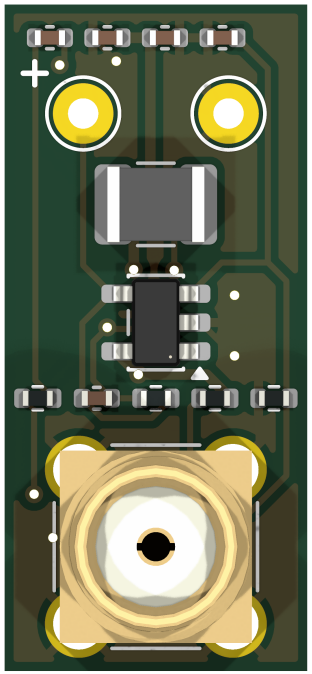
Power the ADC board from 9 V and connect the head to SENSOR IN (J1) with the coax. Measure the head first, then inject a known charge. A 10 pF capacitor from the function generator to the piezo + pad gives Q = 10 pF × ΔV.

A step of 100 mV injects 1 pC. The output moves by 21 mV, because Q / Cf = 1 pC / 47 pF.

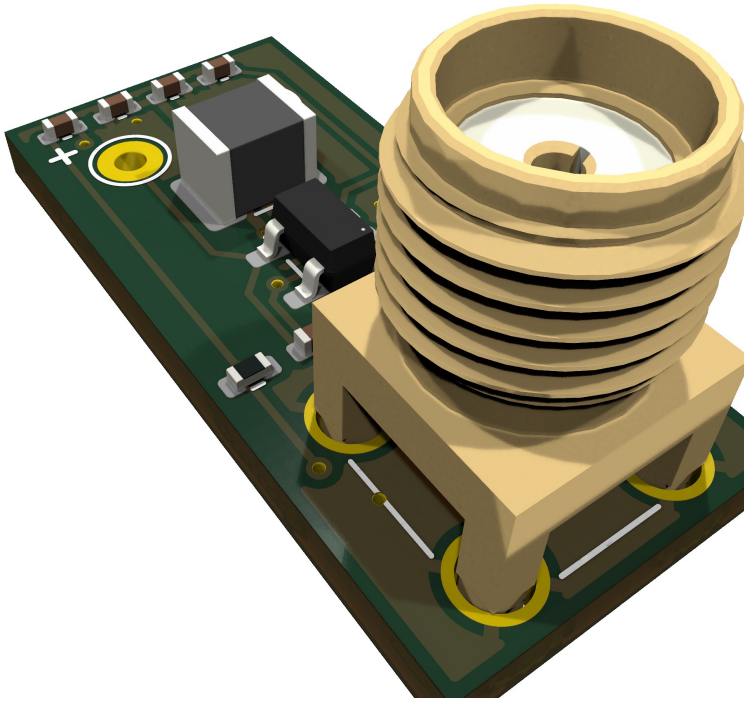
EXPECTED VALUES

Where	Expect
C37 (V+ of U8)	about 4.5 V
VREF, C35	2.50 V after 0.2 s
U8 output, R31 pad 1	VREF, 2.5 V
J1 centre pin	about 4.7 V
Extra current on J2	about 5 mA
100 mV step via 10 pF	21 mV step at SIG_IN

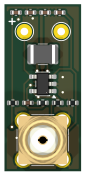
NOTES



Top view at 4 : 1, board 10 × 22 mm



The same board seen at an angle. The piezo goes on the other face, over the two wire holes.



Real size, 1 : 1

WHERE THINGS LIVE

Files, models and datasheets

`hw/`	KiCad project, schematic, both boards
`hw/fab/`	Gerbers, BOM and CPL for JLCPCB, both boards
`hw/tools/`	Generators, DRC, cost model, 1:1 print
`sim/`	Chain simulation, vendor models list, LTspice netlist
`doc/tools/`	This manual, built from the files above
`doc/data/`	Board and simulation snapshots used in the figures

DATASHEETS

ADS7057, THS4531, OPA365 (OPA2365), LM393, XC6206, AMS1117

```
Rebuild: M4_PDF_PYTHON=<python with pymupdf> python3 doc/tools/make_adc_manual.py
```